

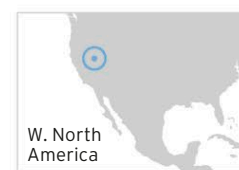


△ YOSEMITE SUNRISE

El Capitan, on the left in this image, is made of granite, while Half Dome, visible slightly right of center, is composed of granodiorite. This variation is a result of different magmas invading different areas of the existing rock.

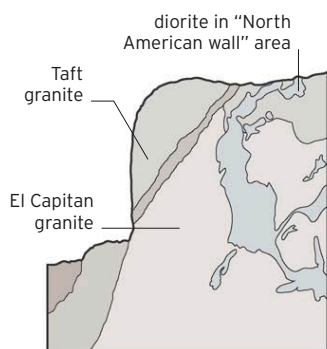
El Capitan and Half Dome

Massive rock outcrops in Yosemite National Park, California, made almost entirely of granite



EL CAPITAN ROCK TYPES

El Capitan is composed mainly of a 100-million-year-old igneous rock known as El Capitan granite. Taft granite, a slightly younger rock, makes up much of the rest.



Tens of millions of years ago, numerous rounded masses of igneous rock, called plutons, formed underground in what is now central California as enormous globs of magma rose from deep in Earth's crust and then cooled. Following several periods of uplift and erosion, many of these plutons are now visible above the surface as massive rock cliffs and outcrops. Two of the most famous, both located in Yosemite National Park, are El Capitan and Half Dome.

Imposing cliffs

El Capitan ("the Chief") is a granitic behemoth that is around 1 mile (1.5km) wide and some 2,950 ft (900m) from base to summit along its tallest face. It was given its name in 1851 by a roving Spanish militia force. Equally impressive, overlooking Yosemite Valley about 5 miles (8km) away,

is Half Dome, named for its distinctive shape. One of its sides is a vertical face around 2,000 ft (600m) high, while the other three sides are smooth and round, making it appear from some angles like a dome cut in half.

The two formations are famous for the challenge they pose to rock climbers. Once considered impossible to climb, El Capitan has now been conquered thousands of times over, and there are more than 70 different routes to the top, of varying difficulty. At Half Dome, an 8-mile (13-km) hiking route has been constructed to the top from the valley floor for non-rock climbers. This includes several hundred steps cut into the granite.



◁ COMEBACK RAPTOR

Peregrine Falcons have been nesting on El Capitan and other sites in Yosemite since 2009, following a 16-year absence.